

Read between the lines : reading comprehension with logic

https://github.com/shreya1297/AppliedML_Group8

Problem

Most banjo players are guitar players.

Most guitar players are not banjo players

Question

Which option must be true if the above statements hold?

Answer Options

A1

A2

A3

A4

Introduction

- ✓ We study how different neural architectures perform on logical reading comprehension by comparing our selected baseline with Simple Average Model, LSTM, and Transformer models trained from scratch.
- ✓ We use hyperparameter tuning, ablations, seed stability, and error analysis to draw robust conclusions.

Research Question

- ✓ Does an LSTM architecture exhibit better performance than a Transformer architecture when both are trained from scratch on a limited dataset (N=4,638)?
- ✓ Sub question: How does this compare to pre trained weights from a Language Model like deberta-v3-large

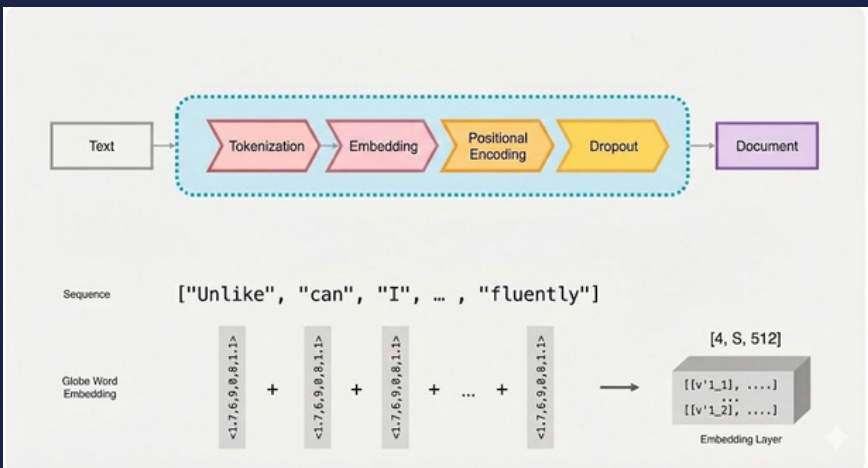
Dataset

- ✓ 4638 Entries
- ✓ 5 Columns
 - id
 - context
 - question
 - answers
 - label

Baseline



Preprocessing



Error Analysis

- ✓ **Failure with Negation:** "EXCEPT" Questions; Example ID: train_1648 (Pesticides)
- ✓ **Task:** Identify the option that does not weaken the argument.
- ✓ **Error:** The model chose an option that weakens the argument (Option 3: Natural pesticides are less potent).
- ✓ **Diagnosis:** The model likely ignored or misunderstood the word "EXCEPT." Small transformers often struggle with negative constraints.
- ✓ **Solution:** Improve model architecture

Computational Analysis and Model Comparison

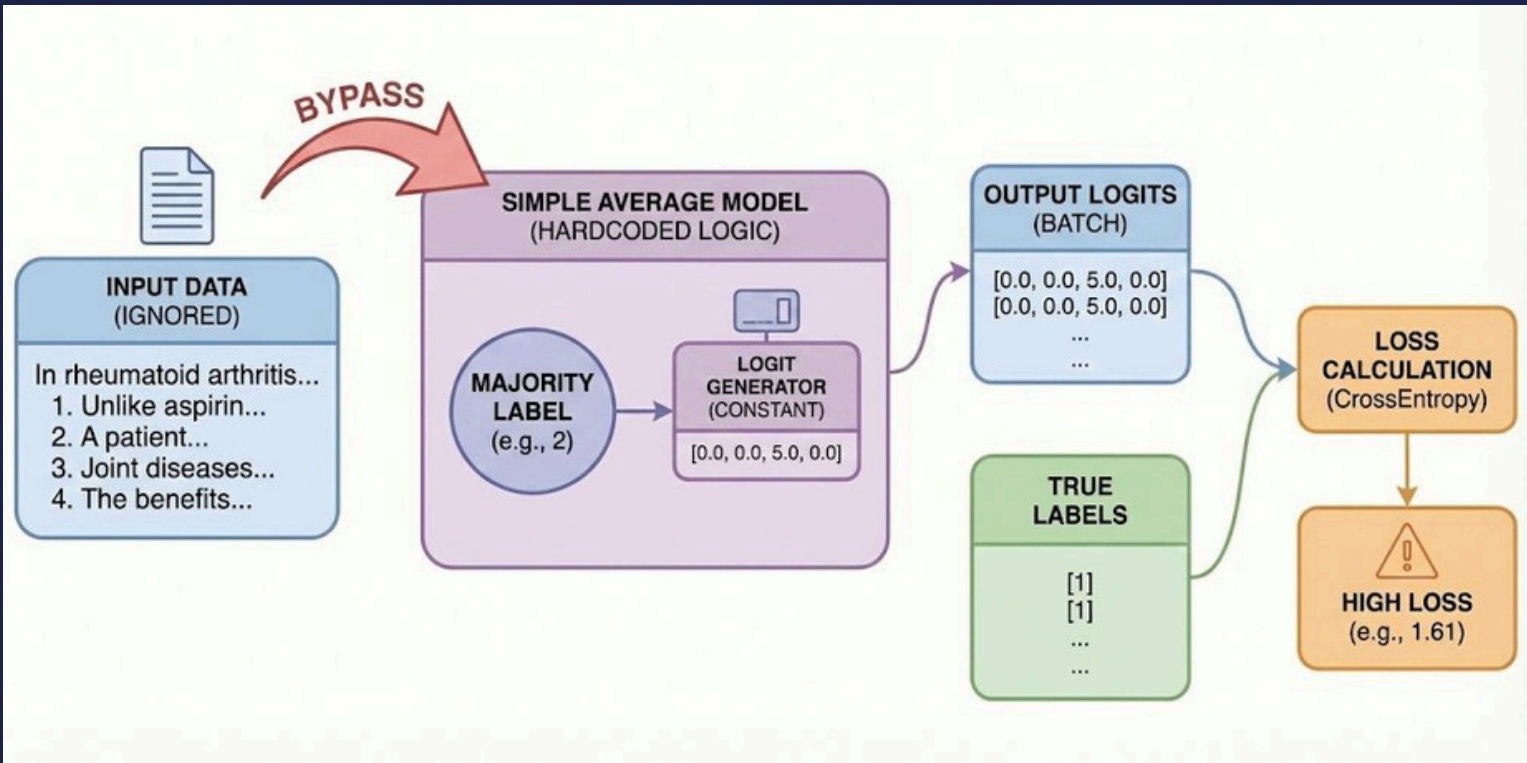
Model	Parameters	Compute Cost	Accuracy
DeBERTa (baseline)	~433M	~53B FLOPs ~1.6 GB memory	0.45
Transformer	~7.15M	~1.34B FLOPs ~27 MB memory	0.358
LSTM	~35.1M	~1.21B FLOPs ~134 MB memory	0.272
Simple Average	~0	~0 FLOPs ~0 MB memory	0.234

Results and Discussion

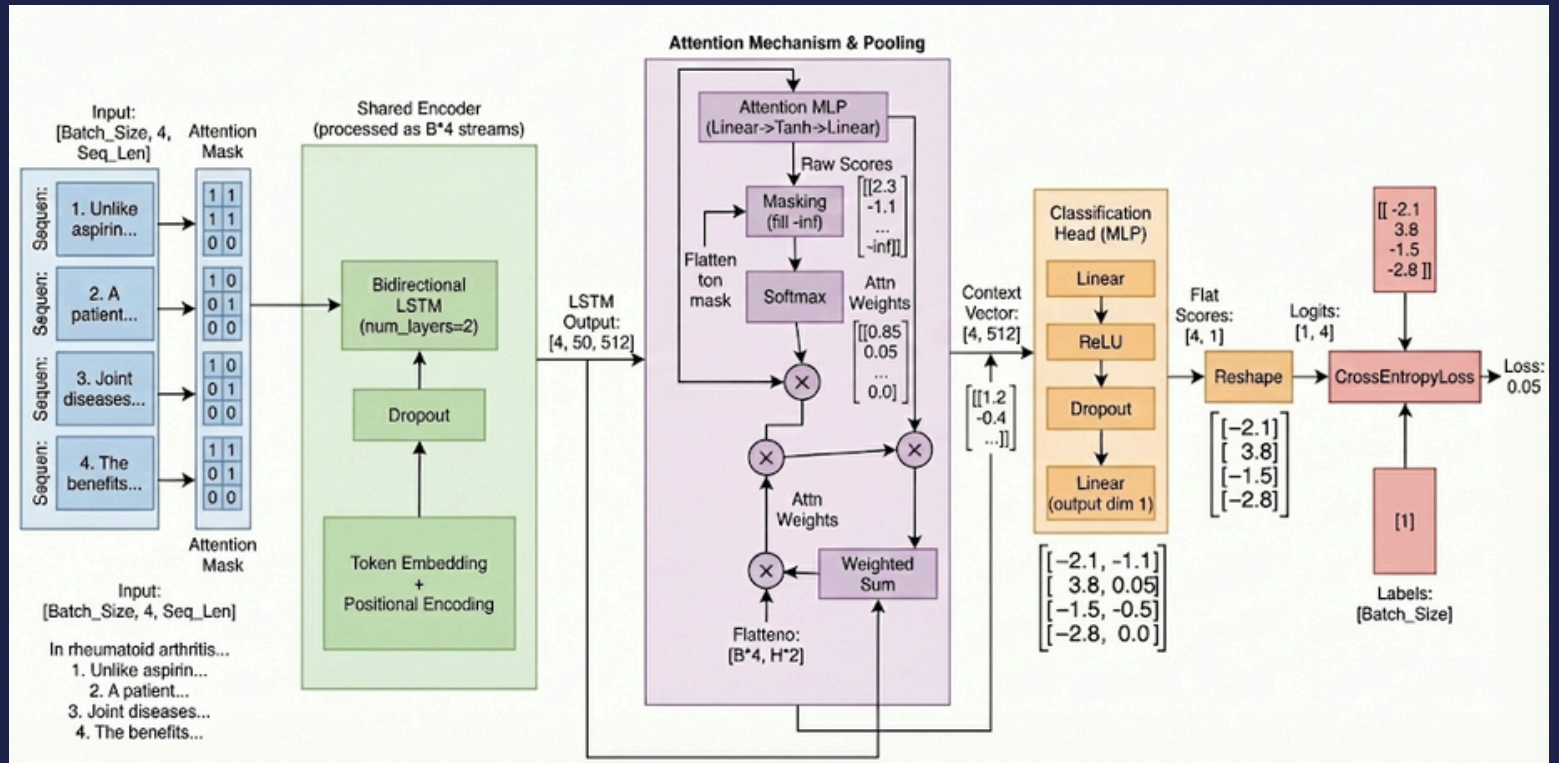
- ✓ DeBERTa (baseline) achieves the highest accuracy, but incurs substantial computational and memory overhead.
- ✓ Transformer (trained from scratch) offers the best trade-off between predictive performance and computational efficiency.
- ✓ LSTM underperforms in accuracy, even when scaled to a high number of parameters.
- ✓ Simple averaging is highly computationally efficient, but yields the lowest accuracy among all approaches.

Model Selection

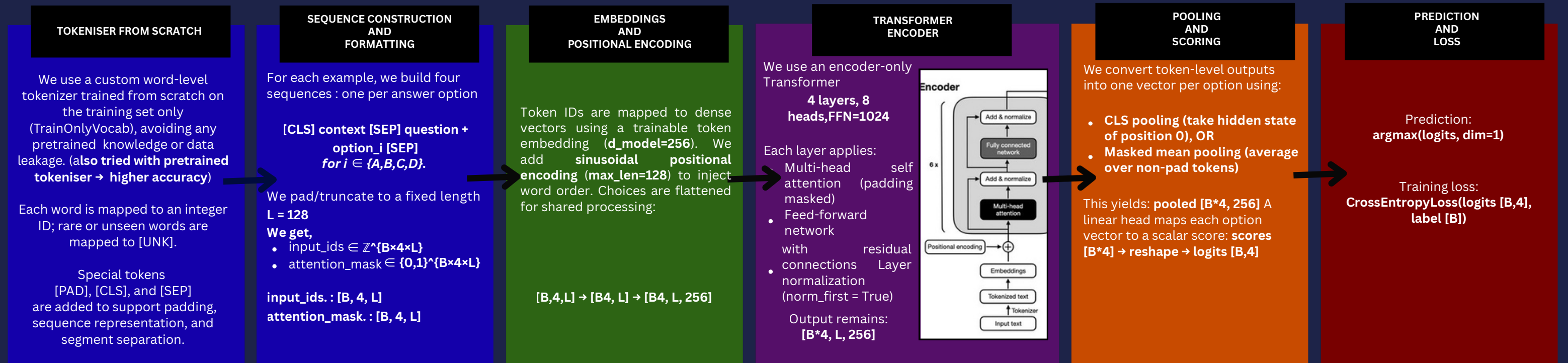
SIMPLE AVERAGE MODEL



LSTM



TRANSFORMER



Conclusion

- ✓ A Transformer trained from scratch offers the best balance between performance and efficiency (Kaggle accuracy: 0.358), while DeBERTa (baseline) achieves **higher accuracy (0.45)** at greater computational cost.